Requirements-based testing of neural networks.

A critical review of RBT4DNN, its empirical evidence, and two focused additional analyses.

CASPAR BRELOH · 17 JULY 2026 / SUPERVISION: VIVEK V. VEKARIYA · SIMON SPETH

Three questions guide the argument.

01 · NEED

Why do neural-network tests need requirements?

PROBLEM + BACKGROUND

02 · METHOD

How does RBT4DNN generate and evaluate targeted tests?

APPROACH + METHODOLOGY + RESULTS

03 · EVIDENCE

How convincing—and scalable—is the resulting evidence?

CRITIQUE + ADDITIONAL ANALYSES

These questions follow the seminar brief while keeping one continuous evidence argument.

High average accuracy does not answer a specific requirement.

CONVENTIONAL TESTING

Known behavior

A specification maps an input condition to an expected output. Test oracles can check the result directly.

DNN TESTING

Learned behavior

The model is inferred from examples. Global accuracy summarizes a distribution, not every meaningful scenario.

THE MISSING LINK

Semantic scenarios

Developers need evidence for statements such as “a very thick 7 must still be classified as 7.”

GLOBAL METRIC
How often is the complete test set correct?

→

SEMANTIC SLICE
Which rare inputs does the score contain?

→

REQUIREMENT VERDICT
Does the model satisfy this specific contract?

[1] §1–§2.5 · “model under test” means the evaluated neural network

Existing generators rarely start from the requirement itself.

Neuron coverage, image transformations, and search-based generators can produce many tests—without proving that those tests represent the scenario a stakeholder cares about.

INPUT	Requirement	Structured natural language defines a semantic precondition and an expected outcome.
METHOD	RBT4DNN	Fine-tune a generator on images that satisfy that precondition.
OUTPUT	Targeted tests	Run the learned component and evaluate the requirement-specific oracle.

[1] §1, §2.5 and §3 · authors claim the first requirement-driven DNN test-input generator

Four domains test whether one workflow survives increasingly ambiguous semantics.

MNIST · 7 REQUIREMENTS

MNIST artifact samples
Source-linked in interactive deck

Digits

Measured height, thickness, and lean. A controlled starting point.

Output: digit class

CELEBA-HQ · 7 REQUIREMENTS

CelebA-HQ artifact samples
Source-linked in interactive deck

Faces

Ambiguous attributes labeled by visual question answering or humans.

Output: eyeglasses decision

SYNTHETIC DRIVING (SGSM) · 7

SGSM artifact samples
Source-linked in interactive deck

Driving

Scene relations linked to steering and acceleration constraints.

Output: numeric control

IMAGENET · 4 REQUIREMENTS

ImageNet artifact samples
Source-linked in interactive deck

Animals

Morphology linked to allowed classes in the WordNet taxonomy.

Output: class set

[1] Table 2 and §4.1.1–§4.1.3 · samples linked from the pinned public artifact

MNIST M3 artifact samples
Source-linked in interactive deck

A requirement defines both relevance and correctness.

If the digit is a 7 and is very thick, then the model shall label it as 7.

PRECONDITION	POSTCONDITION
Which images count as relevant tests?	Which model outputs count as acceptable?

[1] Table 2, M3 · a finite passing suite cannot prove the universal requirement

RBT4DNN turns one requirement into a targeted test campaign.

01 · LABEL

Pixels → terms

Attach domain facts such as very thick, eyeglasses, or same lane.

→

02 · FILTER

Terms → subset

Keep original images satisfying one requirement precondition.

→

03 · ADAPT

Subset → LoRA

Fine-tune FLUX with a small adapter for that precondition.

→

04 · TEST

Images → evidence

Generate tests, run the model under test, and check the required output.

M3 LABELS	FILTERED SUBSET	TARGETED GENERATOR	EVIDENCE
digit = 7 · thickness = very thick	only matching MNIST images	one M3 LoRA adapter	precondition match + output pass or fail

[1] Fig. 2 and §3 · symbols are translated into an explicit input/output contract

Glossary terms make visual requirements operational.

GLOSSARY TERM

A **named visual property** that can be assigned to an image: “very thick,” “same lane,” or “eyeglasses.”

01 · DEFINE

Authors specify the glossary

Requirement concepts are manually turned into a finite vocabulary of visual properties.

TERM CREATION IS NOT AUTOMATIC

02 · LABEL ORIGINAL DATA

Labels come from domain-specific signals

MNIST measurements · SGSM scene graphs · MiniCPM visual Q&A for CelebA-HQ and ImageNet.

CELEBA ALSO HAS HUMAN LABELS

03 · USE THE LABELS

Filter first; check later

Labels select the LoRA training subset. Separate binary classifiers then check glossary terms on generated images.

PRECONDITION MATCH · RQ1 / RQ3

Boundary: separate training labels and evaluation classifiers reduce direct reuse, but neither constitutes independent semantic ground truth.

[1] §3.2 and §4.1.6 · mean glossary-classifier accuracy 94.5%; C4 is 70.8%

One filtered precondition subset trains one targeted LoRA.

FROZEN BACKBONE

W

FLUX.1-dev: a pretrained 12-billion-parameter image generator.

+

TRAINABLE UPDATE

BA

Small low-rank matrices adapt the model to one filtered subset.

BENEFIT	Generate many new candidates inside a rare requirement region
EVIDENCE SOUGHT	Relevance, distributional plausibility, and semantic variation
NOT ESTABLISHED	Non-memorization or systematic coverage within that region

[1] §3.3 and §4.1.4 · LoRA equation simplified; latent-space coverage remains future work

Each baseline answers a different question—compare only within its role.

PRIMARY METHOD

Requirement-specific LoRA

Filtered precondition subset. The central RBT4DNN configuration.

MNIST CONSISTENCY ABLATION

All-data LoRA

Same domain and prompting, but the adapter sees the complete dataset.

DISTRIBUTIONAL QUALITY

Prompt-only FLUX

No domain adaptation. Used for distributional quality comparisons.

MNIST TESTING BASELINES

DeepHyperion + transformations

Established test generators that are not conditioned on the requirement.

Reading rule: compare a method only where it is actually used; the baselines do not answer the same research question.

[1] §4.1.4–§4.1.6 · no truly comparable requirement-targeted baseline existed

LoRA compute is measured; total workflow cost is not.

≈24.7

A100-hours if the 21 primary MNIST, CelebA-HQ, and SGSM adapters run serially.

PER PRECONDITION	LoRA training	52.1–82.5 min
REUSABLE ACROSS REQUIREMENTS	Glossary labels + classifiers	cost not reported
PER GENERATED CAMPAIGN	Sampling + model execution	cost not reported
PER AUDIT SAMPLE	Human validity review	cost not reported

DEFENSIBLE CONCLUSION

Per-precondition LoRA adds roughly linear adapter overhead; the paper cannot tell us whether compute, labeling, engineering, or human validation dominates total cost.

[1] §4.1.4 · derived total excludes ImageNet, extra LoRAs, labeling, generation, evaluation, and engineering

The six RQs validate generated inputs before interpreting model outcomes.

Validate the generated inputs

RQ1–RQ3

RQ1	Consistency	Does the image satisfy the precondition?
RQ2	Plausibility	Is its distribution close to matching training images?
RQ3	Variation	Are other glossary features distributed similarly?

Evaluate the resulting tests

RQ4–RQ6

RQ4	Comparison	Are targeted outcomes more informative than baselines?
RQ5	Violations	Do tests reveal credible postcondition failures?
RQ6	Diagnosis	Do failures expose repeatable decision patterns?

[1] §4.1 · one slide per RQ keeps the evidence and its boundary together

QUESTION Do generated images actually satisfy the requirement precondition?

HOW ANALYZED All required terms must pass separate binary glossary classifiers · 100 images × 10 repetitions.

Across 21 requirements, targeted LoRAs match preconditions 92.1% on average.

MEASUREMENT

One binary glossary classifier per term; mean classifier accuracy: 94.5%.

Domain means differ: CelebA-HQ automated visual-question labels 95.8%; human labels 96.1%; SGSM is weaker at 82.3%.

	Model pass	Precondition match	
M3 · targeted very thick 7	69.4%	95.8%	Most outcomes address the requested scenario.
M3 · all data same prompt	99.6%	7.4%	The excellent pass rate mostly describes ordinary sevens.

MNIST Fig. 3 also includes DeepHyperion and image transformations: neither targets the requirement, and both achieve low match.

[1] §4.2.1 and Fig. 3 · glossary classifiers remain proxies, not ground truth

QUESTION Are generated tests close to real images that satisfy the same precondition?

HOW ANALYZED KID between generated images and the matching filtered subset; targeted LoRA versus prompt-only FLUX · lower is closer.

Targeted LoRA cuts KID by 77.09% for MNIST and SGSM; CelebA-HQ is mixed.

77.09%

average KID reduction across MNIST M1–M6 and SGSM S1–S7

CELEBA · AUTOMATED LABELS	**12.9% average reduction**
CELEBA · HUMAN LABELS	**27% average reduction**
REFERENCE	**matching filtered subset**

Matching training subset · · · · · KID ↓ lower = closer distributions · · · · · Generated tests

Boundary: KID is a distribution-level proxy—not an image-level realism score—and several CelebA-HQ reference subsets contain fewer than 300 images.

[1] §4.2.2 and Table 3 · same prompts; targeted LoRA compared with prompt-only FLUX

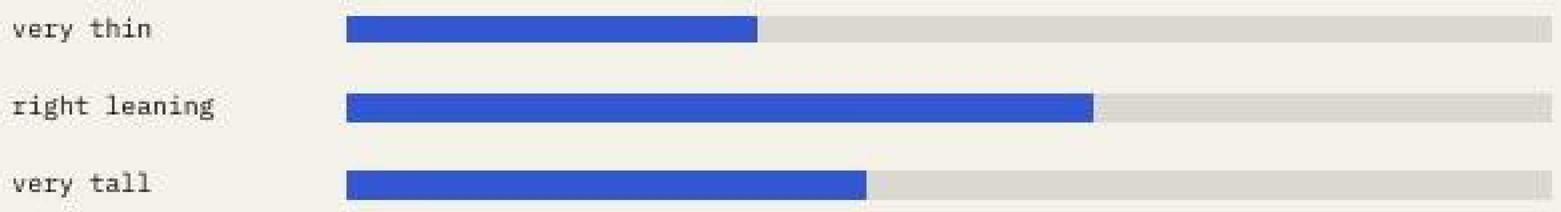
QUESTION
Do generated tests preserve variation in glossary features beyond the targeted term?

HOW ANALYZED
Jensen–Shannon divergence between glossary-term frequency distributions in generated and filtered images · lower is closer.

RQ3 preserves measured feature frequencies—not unrestricted image diversity.

FILTERED TRAINING SUBSET

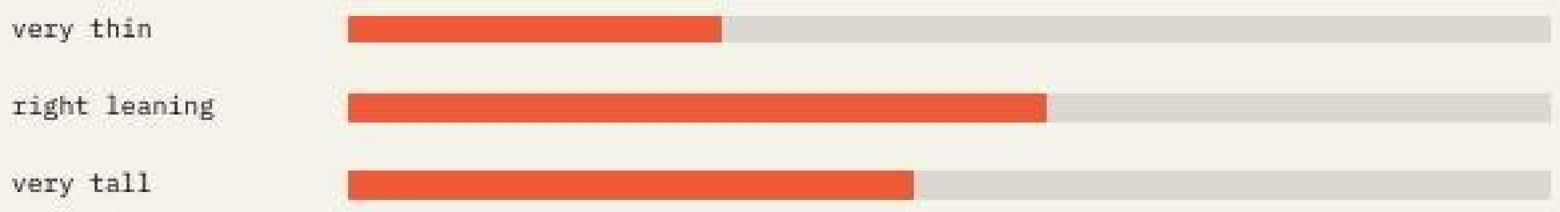
Other glossary features



↔

GENERATED TESTS

Glossary-classifier frequencies



RESULT
MNIST and SGSM targeted-LoRA distributions were more than three times closer on average than prompt-only FLUX; CelebA-HQ improved about 32%.

NOT MEASURED
Duplicates, memorization, pixel diversity, within-term mode collapse, and systematic latent-space coverage.

[1] §4.2.3 and Table 4 · schematic bars explain the metric; they are not reported measurements

QUESTION

Do targeted tests provide more requirement-relevant evidence than existing MNIST baselines?

HOW ANALYZED

For M2 (“very thick 3”), compare precondition match and model outcomes with DeepHyperion and image transformations.

For M2, RBT4DNN reaches the precondition; both untargeted baselines largely do not.

GATE 1 · INPUT

Does this image actually show a very thick 3?

DeepHyperion and transformations generate recognizable digits, but they are not conditioned on M2 and rarely reach its semantic precondition.

→

GATE 2 · OUTPUT

Only then does the classifier outcome address M2.

RBT4DNN trades a superficially higher pass rate for much more requirement-relevant evidence.

Narrow conclusion: requirement-aware generation is a better basis for these MNIST requirements than the two tested untargeted baselines—not universally superior DNN testing.

[1] §4.3.1, Fig. 7 and Table 5 · no truly comparable requirement-targeted baseline was available

QUESTION	Do generated tests expose credible violations of the requirement postcondition?	HOW ANALYZED	Run the model and oracle; then jointly audit 30 failures for each of 11 requirements below 90% pass rate · 330 images.

RBT4DNN found failures for 19 of 21 requirements; reviewed false positives averaged 3.3%.

330

failing images independently assessed by multiple authors and aggregated.

3.3%

mean estimated false-positive rate across the reviewed requirements.

SAMPLE	11 requirements	30 failures reviewed for each requirement whose pass rate was below 90%.
QUALITY	All realistic	Worst estimated false-positive rate was 11.5%; assessor disagreement on precondition consistency was 1.7%.
INSIGHT	Tests expose more than model errors	S2 suggested an underspecified distance condition; S3 exposed generation and labeling difficulty.

[1] §4.3.2 and Fig. 7 · postcondition violations are observed; internal model faults are not localized

QUESTION Can generated failures reveal repeatable patterns in model decisions?

HOW ANALYZED Inspect prediction frequencies and image patterns across 1,486 ImageNet failures · four requirements · three classifiers.

Failures reveal repeatable patterns—but not their causes.

1,486

ImageNet failures examined across the exploratory study.

3

high-accuracy models showed concentrated, model-specific predictions.

OBSERVED **Prediction clusters** Each classifier repeatedly favored particular incorrect labels for some requirements.

ASSOCIATED **Image patterns** Manual inspection found partial animals, background objects, and multi-object scenes.

BOUNDARY **Exploration, not causality** The patterns motivate hypotheses about ImageNet's single-label supervision; they do not prove causal mechanisms.

[1] §4.3.3, Fig. 8 and Tables 6–7 · pass rates span approximately 93.5%–99.5%

The paper demonstrates feasibility—not a complete assurance process.

GENERATIVE TRAINING

Better tests

Improve generator training and adaptation to reveal more observed failures and reduce false positives.

COVERAGE EVIDENCE

Latent coverage

Adapt systematic latent-space coverage when test generation does not reveal a failure.

GENERALIZATION

More domains

Expand driving-like safety datasets and explore broader robustness and functional requirements.

ASSURANCE BOUNDARY

Automated labels, selected datasets, narrow baselines, and small KID subsets limit how far the evidence generalizes.

[1] §4.4 and §5 · author-stated scope kept separate from this seminar's critique

The paper jointly validates 330 sampled failures—but not the full failure population.

PRECONDITION INVALID · MODEL PASSES	PRECONDITION INVALID · MODEL FAILS
Irrelevant pass The output says nothing about the stated requirement.	**Off-target false positive** A model error is not yet a requirement counterexample.
PRECONDITION VALID · MODEL PASSES **Valid requirement pass** Relevant evidence, but never proof of the universal property.	PRECONDITION VALID · MODEL FAILS **Credible counterexample** This joint event is the paper's strongest testing evidence.

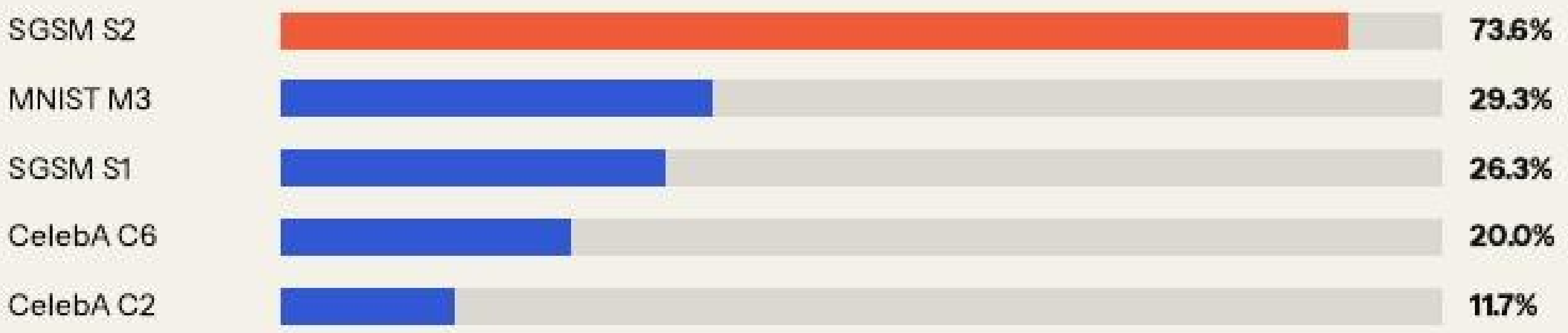
SAMPLED JOINT AUDIT

30 failures × 11 low-pass requirements = 330 independently assessed samples.

Fair critique: the paper did perform joint validation. Its generalization remains limited by author assessment, fixed samples, no reported confidence intervals, and no external or blinded annotators.

The proxy and 42-image LLM check identify audit targets; neither estimates population validity.

Independence-proxy ranking



Diagnostic only · unconditional P(match) × P(fail) · not an observed joint rate

SMALL EXTERNAL SEMANTIC CHECK

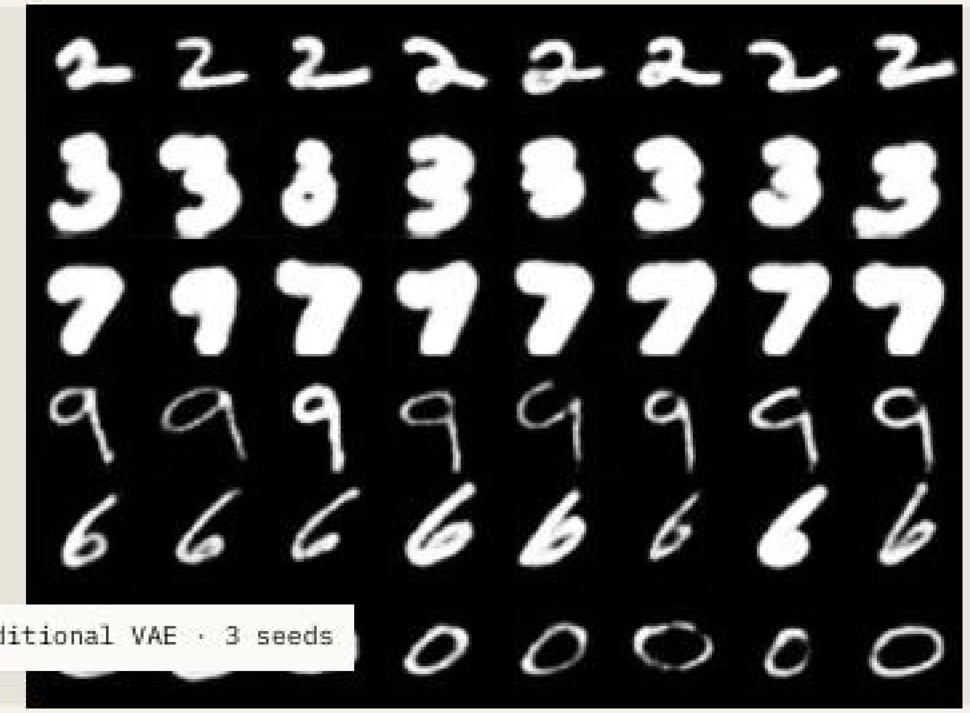
.738

31 of 42 sampled images were judged precondition-valid by one multimodal model.

Use: decide where independent joint review is most valuable.

The product assumes independence. The paper's conditional human audit among failures is methodologically stronger; this LLM check is only a second opinion.

The shared VAE matches MNIST digit-pass rates, but precondition validity was not measured.



MNIST · shared conditional VAE · 3 seeds

.945

mean classifier pass rate

.938

paper v3 repeated-run mean, M1–M6

Weak case	M3 performance	.743
CelebA	generated-image alignment	.613
Evaluator	CelebA evaluator accuracy	.636

Conclusion: the similar classifier pass rate is a promising pilot in a simple domain. Evaluation pipelines differ, and it says nothing yet about precondition validity, realism, diversity, FLUX + LoRA, or natural images.

Requirements make tests relevant. Validity makes failures credible.

WHAT THE PAPER ESTABLISHES	WHAT REMAINS OPEN
Structured requirements can guide realistic, diverse, and scenario-specific neural-network test generation across four domains.	Broader independent joint audits, shared generators, alternative adaptation methods, and end-to-end replication at larger scale.

References.

[1] **N. J. Mozumder, F. Toledo, S. Dola, and M. B. Dwyer.** “RBT4DNN: Requirements-based Testing of Neural Networks.” arXiv:2504.02737, v3, 2025.

[2] **E. J. Hu et al.** “LoRA: Low-Rank Adaptation of Large Language Models.” ICLR, 2022.

[3] **RBT4DNN artifact.** github.com/less-lab-uva/RBT4DNN · seminar provenance pinned to commit c3deff871ed41043d83a05289faab84c41706586.

[4] **Seminar experiments.** Reproducible notebooks, results, summaries, and provenance in this repository.

Original paper ↗	Original artifact ↗